

level 1

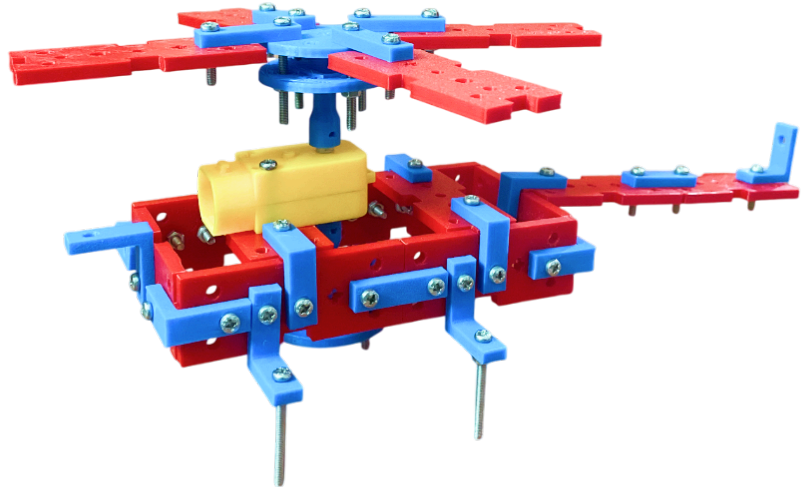
oh.

copter

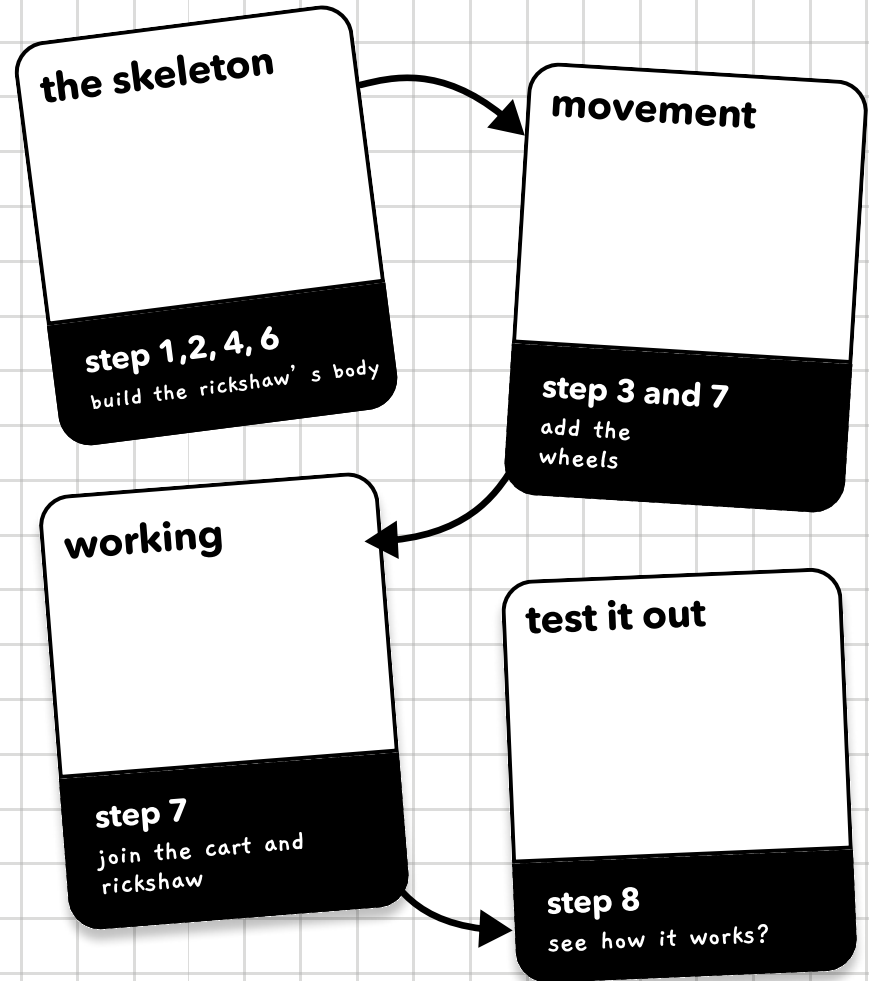
a flying machine powered by gears! When you turn the handle, my gearbox makes the blades spin fast. My gears work together to show how we can use simple machines to create big movements from small turns.

no. of sessions: 6

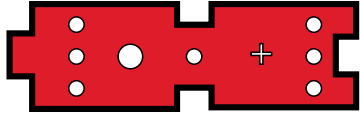
simple machine: gears



what's the goal?



materials needed



m-f beam - x6



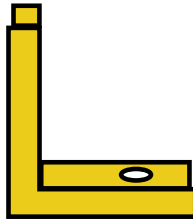
small screw - x12



nut - x12

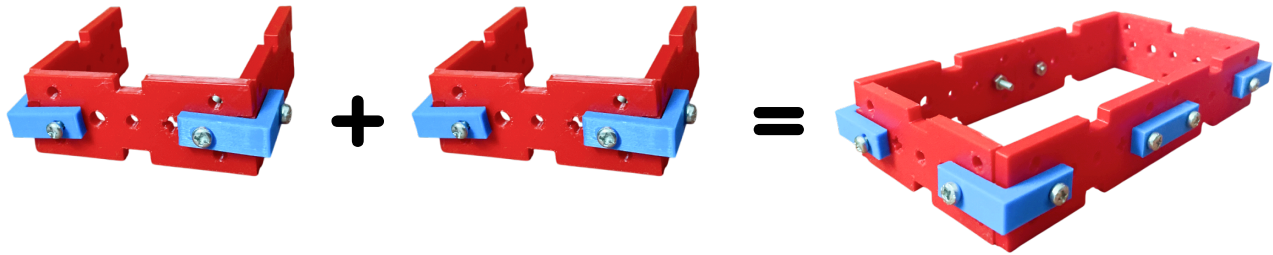


I clamp - x2



L clamp - x4

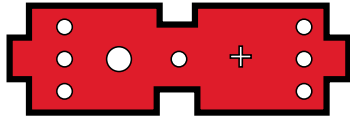
step 1: assemble the chassis



Lets join **3 x M-F beams** to form a U shape.
Use **2 L clamps** to hold them together.

Join the two U shapes
together using **1 clamps**
to form a rectangle.

materials needed



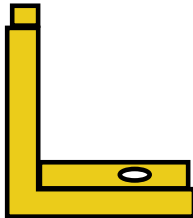
m-m beam - x2



small screw - x4

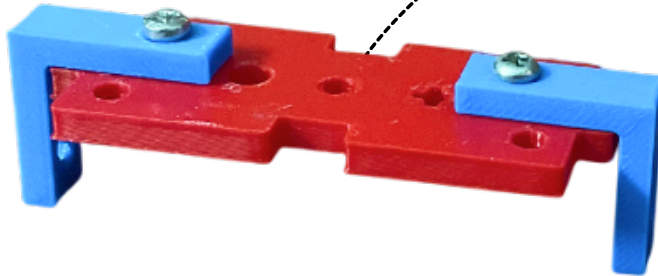


nut - x4



L clamp - x4

step 2: assemble the chassis supports



Use a **M-M beam** to support the chassis.
Add **L clamps** to the ends of the beam as shown.



Repeat this step to make another set.

materials needed

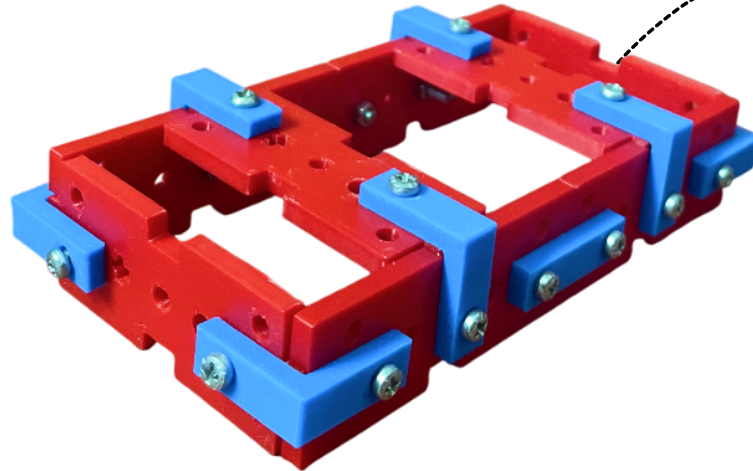


small screw - x4



nut - x4

step 3: complete the chassis assembly



**Join the two
supports to the
chassis as shown**

materials needed



coupler - x1



gear box x1



long screw - x2

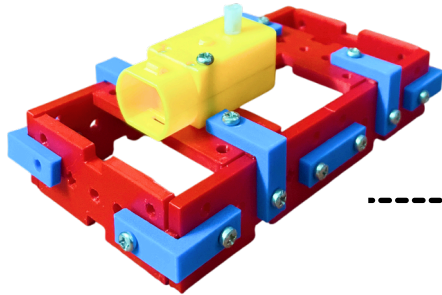


nut - x2



Do not tighten the gear box too much, keep it slightly loose so the gears inside can move easily.

step 4: assemble the gear box

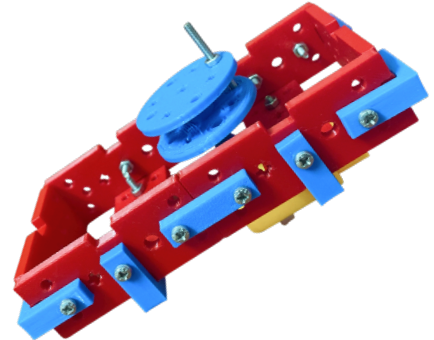


Attach a gear box to the center of any one beam as shown.

use a long screw to hold them together.

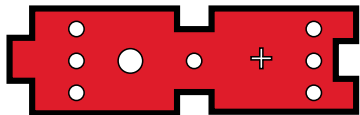


Take a coupler, pass a long screw through it and add nut



Fix the coupler to the axle of the gear box as shown.

materials needed



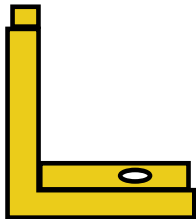
m-f beam - x2



small screw - x4

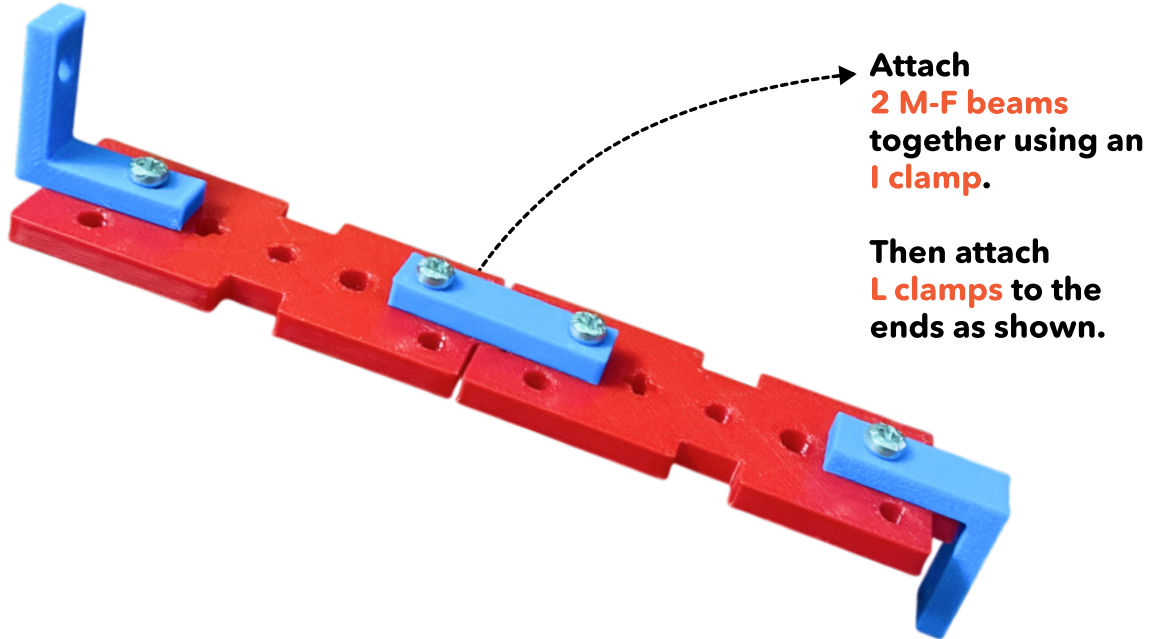


nut - x4



L clamp - x2

step 5: assemble the copters tail



materials needed



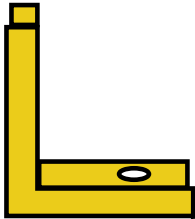
long screw - x4



small screw - x6



nut - x10

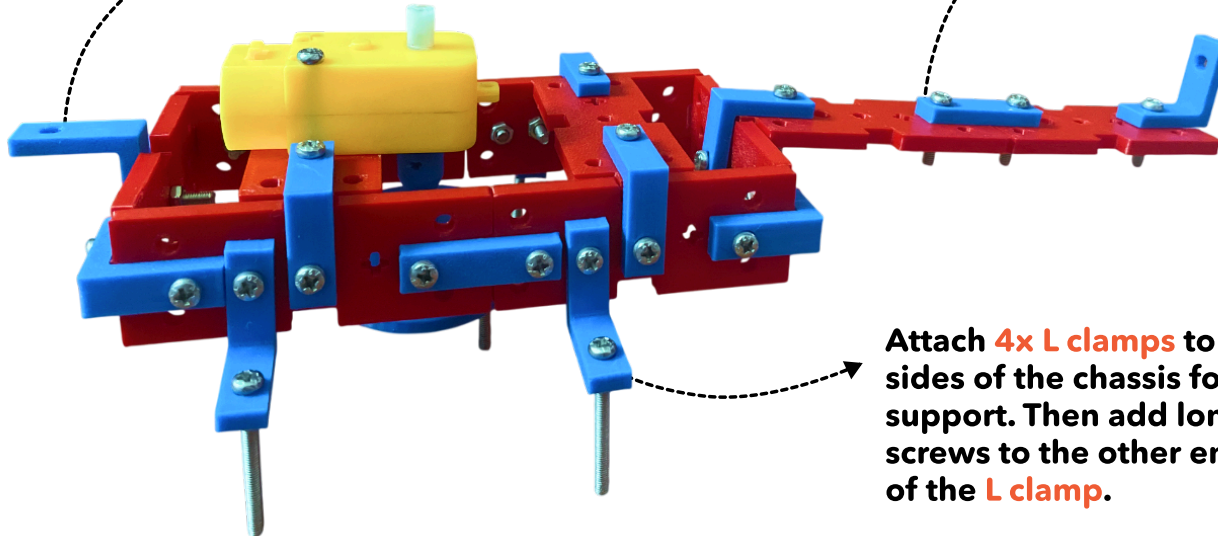


L clamp - x5

step 6: assemble the tails & copter supports

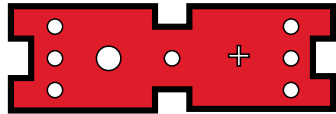
Fix **1 L clamp** to the front the chassis as shown.

Fix the tail to the back of the chassis as shown



Attach **4x L clamps** to the sides of the chassis for support. Then add long screws to the other ends of the **L clamp**.

materials needed



f-f beam - x4



long screw - x4



small screw - x4



nut - x8



coupler - x1

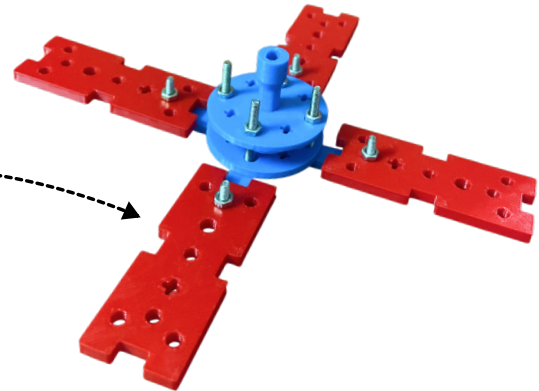


I clamp - x4

step 7: assemble the blades

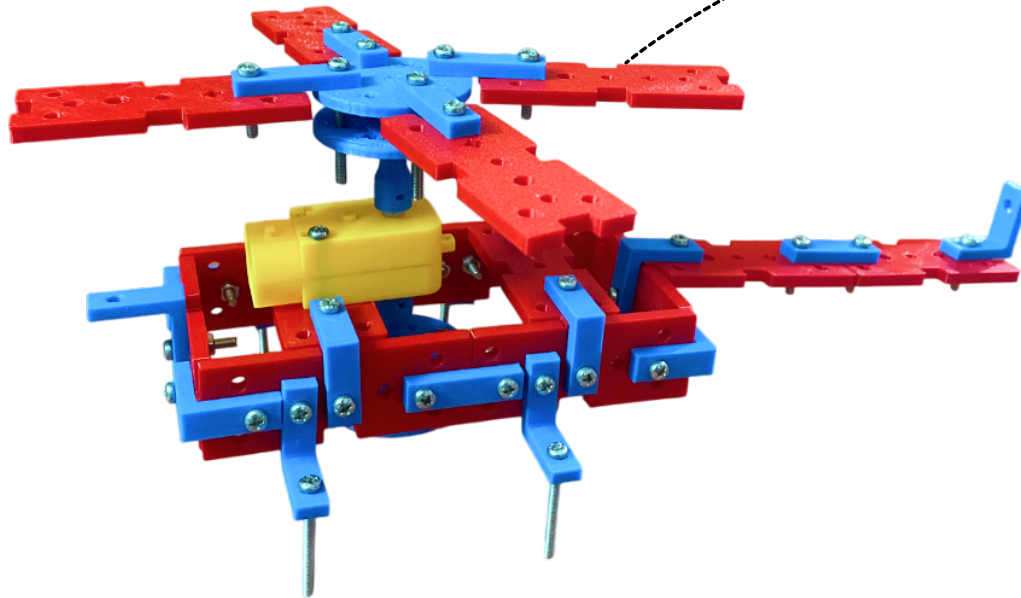


Join **4x I clamps** to the holes on the **coupler** as shown. Use **long screws** to hold them together.



Add **F-F beams** to the ends of the **I clamp** as shown.

step 8: complete the blade assembly



Fix the blades to the other end of the motors axle as shown.

**With this step
your copter is
all set!**